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End Semester Examination December, 2025

Name of the Course: ECE
Name of the Paper: Digital Electronics
Time: 3 Hours

Semester: IIIrd
Paper Code: TEC 302
Maximum Marks: 100

Note:-

- (i) All questions are compulsory.
- (ii) Answer any two sub questions among a, b & c in each main question
- (iii) Total marks in each main question are twenty.
- (iv) Each question carries 10 marks

Q1	(20marks)	
(a)	Perform following: i) $(456)_8 + (757)_8 = (\quad)_{16}$ ii) 14 - 19 using 2's complement.	CO1
(b)	Implement the following expression using 2:4 Decoder $F(A,B,C) = A'BC + AB' + AC'$	
(c)	Solve the following expression using Quine McCluskey method $Y = \sum m(1,2,3,5,7,10,11,14) + d(0,8,9,15)$.	
Q2	(20 marks)	
(a)	Design a BCD to EXCESS-3 code converter.	CO2
(b)	Implement the expression $F(A,B,C) = A'B + AB' + C'$ using 4:1 multiplexer.	
(c)	Design a 4 bit parallel adder and subtractor.	
Q3	(20 marks)	
(a)	What are the limitations of level triggered circuit? Explain Master Slave Flip flop.	CO3
(b)	Explain working of universal shift register with suitable diagram.	
(c)	What are the limitations of Asynchronous counter? Design a synchronous counter for sequence $0 \rightarrow 1 \rightarrow 3 \rightarrow 5 \rightarrow 0$ using JK Flip Flop.	
Q4	(20 marks)	
(a)	What are Registers? Explain 4-bit PISO shift register.	CO4, CO6
(b)	A Synchronous sequential logic circuit with input x is described by the following function $T_A = x \text{ AND } Q_B$ $T_B = x \text{ OR } Q_A$ Analyze the circuit accordingly.	
(c)	Design a 2-bit counter with external input p. The counter counts up when input is 1 otherwise it counts down.	
Q5	(20 marks)	
(a)	Explain with neat diagrams CMOS NOR Gate.	CO5
(b)	Explain operation of ECL AND Gate	
(c)	Classify logic families. Explain different characteristics of Logic families with example.	

